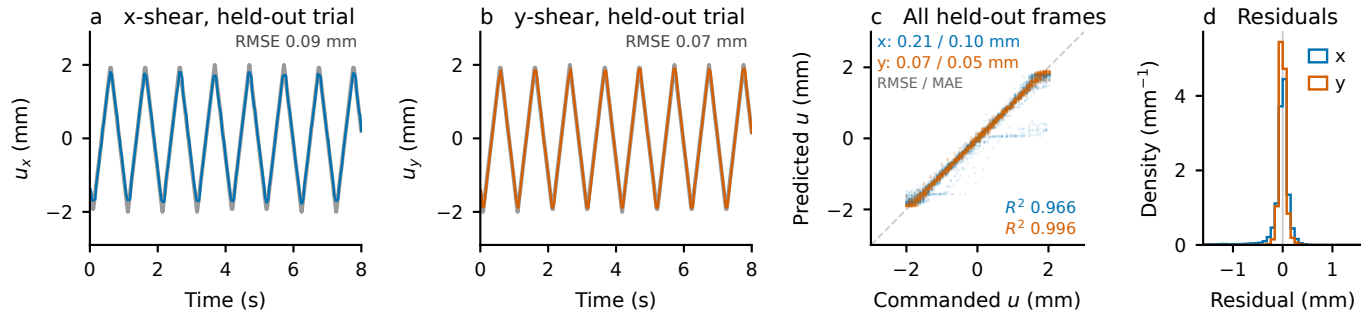


— commanded u (no encoder readback) — predicted $\hat{u}_x$ (LOTO) — predicted $\hat{u}_y$ (LOTO)



Leave-one-trial-out over 17 x- and 18 y-trials; random forest shown (ridge: x 0.19 mm, y 0.11 mm RMSE; mean-prediction baseline 1.16 mm). Truth is the commanded platen trajectory (no encoder readback); constant speed between magnetically anchored extremes assumed. Flat-platen shear-displacement decoding, not point-contact localization.